

DAC Clocking and I2S Interface (ES9038Q2M): Working Document

Digital Audio Player • Rev 10

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

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1. Summary

Field	Value
Manufacturer / MPN	ESS Technology, ES9038Q2M (#1-053), B100
Package	QFN with exposed pad (exposed pad = DGND)
Function	32-bit stereo audio DAC, differential current outputs
Timing role	I2S master; the ESP32-S3 is I2S slave
Clock sources	Two tri-state oscillators on XI: 22.5792 MHz (Y100) and 24.576 MHz (Y101)
Analog supply	AVCC_L, AVCC_R and VCCA on +3V3_ADC, a dedicated LT3042

DPLL off (synchronous mode): every audio clock derives from one local oscillator. The output scales with AVCC ($V_{OPP} = 0.906 \times AVCC$), so AVCC has its own regulator and carries no op-amp current.

Source: [1], [5]

2. Design Requirements

Requirement	Value
Input format	PCM over I2S. No DSD, no S/PDIF.
Sample rates	44.1, 88.2, 176.4 kHz and 48, 96, 192 kHz
Timing roles	DAC master (Reg 10 master_mode = 1)
DPLL	Off: MCLK = $128 \times f_S$
Clocking	One of two oscillators enabled at a time
I2C speed	100 kHz ($SCL < MCLK / 20$)
Clock switching	Only while muted or RESETB is low
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	AVCC_R	+3V3_ADC, 0.1 μ F + 1 μ F to GND (C114, C115)
2	DACRB	Right negative current output, to I/V (U100 -IN A)
3	DACR	Right positive current output, to I/V (U100 -IN B)
4 to 7	BIAS, SW, FSYNC, HPSPDb	Not connected
8	DACL	Left positive current output, to I/V (U101 -IN B)
9	DACLB	Left negative current output, to I/V (U101 -IN A)
10	AVCC_L	+3V3_ADC, 0.1 μ F + 1 μ F to GND (C116, C139)
11	AGND_L	GND
12	VCCA	+3V3_ADC, 0.1 μ F + 1 μ F to GND (C119, C120)
13	DVCC	+3V3_DIG, 1 μ F + 0.1 μ F to GND (C121, C122)
14	DGND	GND
15	DVDD	4.7 μ F to GND only (C117)
16, 17	GPIO2, GPIO1	GPIO2 / GPIO1 of the ESP32-S3, 10 k Ω pull-downs (R113, R112)
18	DATA2	PCM data input from GPIO9
19	DATA1	LRCLK output to GPIO8
20	DATA_CLK	BCLK output to GPIO7
21	RESETB	GPIO36, 10 k Ω pull-down (R111)
22	RT1	GND (reserved)
23, 24	SCL, SDA	GPIO5 / GPIO4, 10 k Ω pull-ups to +3V3_DIG (R110, R109)
25	ADDR	10 k Ω pull-up to +3V3_DIG (R108): 0x92 write, 0x49 (7-bit)
26	NC	Not connected
27	VCCA	+3V3_ADC, 0.1 μ F + 1 μ F to GND (C112, C113)
28	XOUT	Not connected

Pin	Name	Connection
29	XI	Oscillator outputs, each through 22 Ω (R104, R105)
30	AGND_R	GND
EP	Exposed pad	DGND (GND)

Source: [1], [4]

GPIO1 / GPIO2 default mode 13 shuts the DAC down when high; both carry pull-downs (Register 8, datasheet p. 22).

4. External Components

Ref	Value	Part / Notes	AL-PN
X1 (Y100)	22.5792 MHz	ECS-2520MVL-225.792-CN-TR, XO ± 25 ppm, tri-state OE	#1-100
X2 (Y101)	24.576 MHz	Abracon ASTX-H11-24.576MHZ-T, TCXO ± 2.5 ppm, tri-state OE	#1-101
R_S1, R_S2 (R104, R105)	22 Ω 1 % 0402	YAGEO RC0402FR-0722RL	#1-102
R_EN1, R_EN2 (R106, R107)	10 k Ω 0402	OE pull-downs; YAGEO AC0402FR-0710KL	#1-134
R_RST, R_GPIO1, R_GPIO2 (R111 to R113)	10 k Ω 0402	Pull-downs	#1-134
R_SCL, R_SDA, R_ADDR (R110, R109, R108)	10 k Ω 0402	Pull-ups to +3V3_DIG	#1-134
C_DVDD (C117)	4.7 μ F 6.3 V X5R 0402	Murata GRM155R60J475ME47D	#1-103
Decoupling, 0.1 μ F	0.1 μ F X7R 0402	Each AVCC, VCCA, DVCC and oscillator VDD (C103, C104, C112, C114, C116, C119, C122)	#1-080
Decoupling, 1 μ F	1 μ F 25 V X7R 0603	Each AVCC, VCCA, DVCC and oscillator VDD (C102, C105, C113, C115, C139, C120, C121)	#1-132

Source: [2], [3], [4]

Oscillator enable: pin 1 low = output high impedance. Pinout of both oscillators: 1 OE, 2 GND, 3 OUT, 4 VDD. Oscillators stay on +3V3_ANALOG so their clock-edge currents stay off +3V3_ADC.

5. Equations and Results

$$f_{MCLK} = f_{XI} / 2^{clk_gear}; \quad f_{MCLK} = 128 f_S; \quad f_{BCLK} = f_{MCLK} / 2 = 64 f_S \quad (1)$$

$$f_{SCL} < f_{MCLK} / 20 = 5.6448 \text{ MHz} / 20 = 282 \text{ kHz} \quad (2)$$

Source: [1], pp. 14, 15, 24, 52.

Oscillator	clk_gear	f_S (kHz)	MCLK (MHz)	BCLK (MHz)
22.5792 MHz	$\div 1 / \div 2 / \div 4$	176.4 / 88.2 / 44.1	22.5792 / 11.2896 / 5.6448	11.2896 / 5.6448 / 2.8224
24.576 MHz	$\div 1 / \div 2 / \div 4$	192 / 96 / 48	24.576 / 12.288 / 6.144	12.288 / 6.144 / 3.072

Firmware: hold RESETB low, enable one oscillator, wait for start-up ($ECS \leq 5$ ms), release RESETB, configure over I2C.

6. Important Values

Item	Value
X1 phase noise	-159.8 dBc/Hz at 100 kHz offset
X2 phase noise	-158 dBc/Hz at 100 kHz offset
MCLK range	5.6448 to 24.576 MHz
Maximum f _S , synchronous mode	768 kHz (DAC limit); 192 kHz by oscillator choice
I2C address	0x92 write / 0x49 (7-bit), ADDR high
+3V3_ADC load	≈ 8 mA (AVCC 6 mA, VCCA 2 mA)
Power-up state	OE lines, RESETB and GPIO1/2 pulled low: oscillators off, DAC in reset

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0 to 7	2026-09-23	Interface, clocking, oscillators, pull-downs, 22 Ω series.
8, 9	2026-09-25, 26	Reformatted; maximum sample rate 176.4 / 192 kHz.
10	2026-10-03	AVCC_L, AVCC_R and VCCA moved to +3V3_ADC (dedicated LT3042). Decoupling recorded as fitted: 1 μF X7R 0603 (#1-132) on each supply and oscillator, DVCC 1 μF + 0.1 μF. GPIO and designator mapping added; 7-bit address 0x49.

9. Sources

1. ES9038Q2M Datasheet v1.4 (ESS Technology)
2. ECS-2520MVLC Datasheet (ECS Inc.)
3. ASTX-H11 Datasheet (Abracon)
4. DAP schematic sheet DAP-Audio-1, DAC (author, 2026-10-03)
5. Positive LDO (LT3042), Rev 7